

3 (long) The amplifier with a depletion mode MOSFET in Fig. 3 is biased to achieve $V_{GS} = -3$ V, $V_{DS} = 8$ V and $I_D = 3$ mA. The gate resistor R_2 is $10\text{ M}\Omega$. The supply voltage V_{DD} is 20 V. The FET has small-signal parameters $g_m = 6\text{ mA/V}$ and $r_d = 10\text{ k}\Omega$. It can be assumed that the FET is ideal and that there is no gate current.

- (a) Briefly explain how the FET is biased. [4]
- (b) Find the values of R_1 and R_3 to set the desired operating point. [4]
- (c) Draw the small-signal model of the circuit, assuming that the reactance of the capacitors are negligible. Hence derive the expressions for small-signal gain and output and input impedances. [10]
- (d) The low frequency -3 dB point of the amplifier is dominated by C_{out} and is 20 Hz when a load resistor of $5\text{ k}\Omega$ is connected to the output. Determine the value of C_{out} . You may assume that the reactance of C_s is small compared to R_3 . [8]
- (e) If C_{out} remains unchanged, what load resistor value is needed to maximise the signal power to the load? Determine the new low frequency -3 dB point with this load resistor. [4]

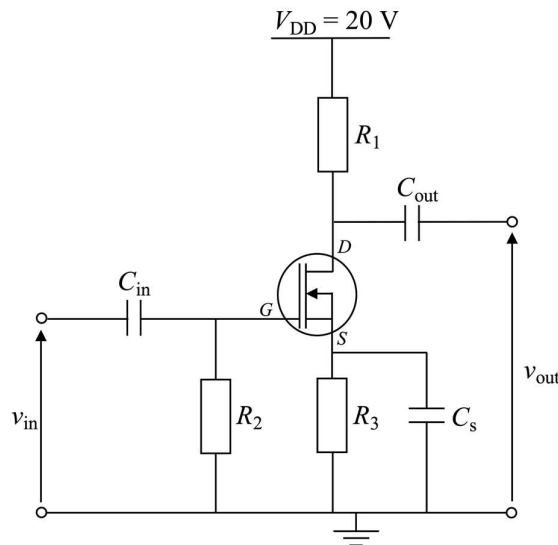


Fig. 3